Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Forward Deployed Engineer with 2+ years turning ambiguous business workflows into production **LLM** applications, **RAG**, and automation in **Python**, with fallbacks, monitoring, and access controls, and measurable impact like 60% faster turnaround.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Web Technologies, Database Systems, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Retrieval, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Engineer *Los Angeles, CA*
- Shipped a multi-tenant AI education platform on Azure SQL and Blob Storage with **row-level security** protecting data privacy for 500+ users, cutting dashboard latency from 1.2s to 350ms.
- Productionized **LLM orchestration** with FastAPI background workers, explicit job states, and failure isolation, processing 1K+ AI jobs at 3x throughput while monitored pipelines held 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*
- Built an **LLM application** that automates grading with prompt chains for rubric generation, OCR, scoring, and feedback across GPT-4o, Claude, and Qwen-VL, cutting turnaround 60% for 150+ instructors.
- Added **fallback paths** that rerouted OCR to Claude Vision on timeouts, logged per-call tokens and cost per feature, and sustained 90%+ grading consistency across 10K+ daily requests.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*
- Automated cross-platform data-sharing workflows that cut manual data entry 35%, partnering with Microsoft and GitHub **stakeholders** in Agile sprints from requirement analysis through deployment and production support.
- Built 20+ unit and integration **test** suites in nightly Jenkins CI/CD with emailed coverage and failure reports, and led an AWS Glue and S3 ETL redesign boosting pipeline efficiency 68%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*
- Used Google **Analytics** on feature usage, load times, and drop-off points to guide product changes that cut resource wastage 42% and raised lead conversions 15% across CRM workflows.
- Built a MERN CRM with eight microservices including JWT authentication and **role-based access** services, monitored via Sentry, Azure Monitor, and Kibana, syncing 2M+ records across locations in real time.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*
- Gathered **client requirements**, mapped 12+ integration points, and presented the end-to-end architecture for sign-off before building Python booking and payment APIs with sub-200ms latency across 50K+ records.
- Implemented passwordless wallet-signature **authentication** and fraud-checked multi-currency payment webhooks for refunds, disputes, and chargebacks, eliminating password storage for wallet users and cutting manual payment handling by 70%.

Skills

Languages: Python, TypeScript, JavaScript, SQL, Java
AI Engineering: LLM Applications, RAG, Embeddings, Prompt Engineering, Prompt Chaining, Multi-Model Routing, Model Fallbacks, OCR, Claude, GPT-4o
Backend & Data: FastAPI, Flask, Node.js, REST APIs, PostgreSQL, SQL Server, MongoDB, ETL Pipelines, Databricks
Cloud & Operations: AWS, Azure, GCP, Docker, CI/CD, Sentry, Azure Monitor, Logging and Monitoring
Security & Governance: Row-Level Security, RBAC, JWT, Snyk, Authentication, Data Privacy
Business: Requirements Gathering, Stakeholder Communication, Impact Measurement, Agile/Scrum

Projects

Guardrailed Support Agent | *Python, FastAPI, Claude, pgvector, PostgreSQL, Docker*
- Designed a Python support agent using RAG over policy docs that calls account tools only after **identity verification**, redacts PII, escalates to humans, and ships behind **evals** for accuracy and containment.

Big-Brain Live | *LLaMA-2, FAISS, Clerk, Convex, PostgreSQL*
- Built and deployed a LLaMA-2 **RAG** assistant with FAISS retrieval, summarization, and Clerk-secured document access, cutting document review time 60% for 200+ users at 95% query accuracy.

Comprehensive Analysis of Blockchain for User Authentication Paper | *RBAC, Self-Sovereign Identity*
- Co-authored an IEEE paper evaluating blockchain-based **user authentication**, IoT device identity, and role-based access with self-sovereign identity, analyzing identity theft, replay, man-in-the-middle, and phishing risks to centralized verification.